

## **BACKLOG BUDDY ? STUDY MATERIAL**

Engineering Mathematics-I ? Unit-wise Lecture Notes

Subject Code: MA101 Semester: 1

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Condensed notes covering all five units: Matrices and Linear Algebra, Differential Calculus, Integral Calculus, Vector Calculus, Ordinary Differential Equations.

### **Unit 1: Matrices and Linear Algebra**

- Definition, key concepts and significance of Matrices and Linear Algebra
- Important diagrams / derivations / algorithms for Matrices and Linear Algebra
- Solved examples and previous-year questions on Matrices and Linear Algebra

### **Unit 2: Differential Calculus**

- Definition, key concepts and significance of Differential Calculus
- Important diagrams / derivations / algorithms for Differential Calculus
- Solved examples and previous-year questions on Differential Calculus

### **Unit 3: Integral Calculus**

- Definition, key concepts and significance of Integral Calculus
- Important diagrams / derivations / algorithms for Integral Calculus
- Solved examples and previous-year questions on Integral Calculus

### **Unit 4: Vector Calculus**

- Definition, key concepts and significance of Vector Calculus
- Important diagrams / derivations / algorithms for Vector Calculus
- Solved examples and previous-year questions on Vector Calculus

### **Unit 5: Ordinary Differential Equations**

- Definition, key concepts and significance of Ordinary Differential Equations
- Important diagrams / derivations / algorithms for Ordinary Differential Equations
- Solved examples and previous-year questions on Ordinary Differential Equations

Exam Tips:

- \* Start with the latest syllabus and exam pattern ? focus on units with the highest weightage first.
- \* Solve at least 3 previous year papers under timed conditions before the exam.
- \* Focus on frequently repeated topics ? they are strong preparation priorities based on available papers (not guarantees).
- \* Revise definitions, diagrams and standard derivations ? they earn easy marks.
- \* Practise numerical problems from every unit; papers usually include 2?3 compulsory problems.

**Disclaimer: Sample study material for preparation support only.**

Appearance counts are based on available papers and are not exam predictions.